

Brad's YANG Description Quality & Definitional Anchoring study — reading memo

Prepared for: Chris **Date:** 22 September 2026 **Source:** Brad's "YANG Description Quality & Definitional Anchoring, Findings" (19 pp.) **Purpose:** Make the study legible, reconcile it with the reconciliation framework, separate what is solid from what is soft, and mark where it can strengthen the master report and a possible NMRG Internet-Draft.

1. What the study is, in one paragraph

Brad is asking a question underneath our whole line of work: when two models describe the same networking concepts in different words, is the *meaning text* attached to a node (its YANG **description**) actually good enough, and actually necessary, for a machine to recognise what that node is? He answers it in two phases. **Phase 1** audits the quality of YANG description statements across seven modules (IETF and TAPI) — are they real definitions or filler? **Phase 2** runs a blind experiment: it makes an LLM bind YANG nodes to concepts in a small shared lexicon, once using only the node's *name and synonyms*, once using its *description text*, and measures which does better. The name-only-vs-definition contrast is a direct proxy for "do surface labels carry the meaning, or does the description?" — which is the empirical heart of the portability and reconciliation argument.

The two corpora:

- **IETF:** 4 modules (**ietf-network** , **ietf-network-topology** , **ietf-te-topology** , **ietf-otn-topology**) — 2,279 scored data-nodes.
- **TAPI:** 3 modules (**tapi-topology** , **tapi-connectivity** , **tapi-common**) v2.5.2 — 3,418 scored data-nodes.

2. Phase 1 — how good are the descriptions?

2.1 The rubric and the headline number

Descriptions were scored against an **ISO-704** definitional-quality rubric on a stratified 150-node sample (75 per corpus): is a description **intensional** (a genuine definition = a genus plus a differentiating characteristic), or does it fail as **tautological/name-restating**, **extensional-example-only**, or **empty/boilerplate**?

Only ~44–45% of descriptions in either corpus are genuine definitions, and the failure modes differ by corpus:

Corpus	intensional	tautological	example-only	empty/boilerplate
IETF	44.0%	49.3%	6.7%	0.0%
TAPI	45.3%	32.0%	0.0%	22.7%

So IETF's characteristic failure is **tautology** — restating the node's own name (~49%) — while TAPI's is **literal boilerplate**, mostly the literal string **"none"** (~23–31%, a leftover from TAPI's UML → YANG code generation).

The nuance Brad draws out: this refutes a naive "IETF good, TAPI bad" prior. When TAPI *does* define, its descriptions are if anything *slightly more* intensional than IETF's; TAPI just has more that are outright blank.

Caveat on the number: both the "manual" and "LLM" annotation passes were produced by the same model (Claude) reading the text twice, so the reported Cohen's $\kappa = 0.880$ measures *rubric self-consistency*, not agreement with an independent human. Treat it as a reliability floor, not validation.

2.2 Duplicate descriptions — the concept is identical, the reuse site is not

There are **429 groups of byte-identical descriptions**, because YANG groupings are reused across many structurally-repeated points (one IETF string appears 136 times; TAPI's generic "The specific value." / "The specific unit of measurement of the capacity." patterns appear 60–120+ times each). This is *not* per se a defect — the concept genuinely is the same everywhere the grouping is reused. But it means **description text alone often cannot distinguish two different reuse sites of the same grouping**.

2.3 Same-corpus semantic collisions — an instance-level limit, named precisely

Within IETF, 214 node pairs have near-identical descriptions (TF-IDF cosine > 0.85) but sit at different paths/scopes — same grouping content, different location. The cleanest example: two nodes, one under **optimizations** in a configured connectivity-matrix and one under **information-source-entry**, carry the *exact same* description ("The objective function container that includes attributes to impose when computing a TE path.") but are legitimately different things (different scope, lifecycle, potentially different values). **The description was never going to distinguish them, because it is the same string in both places by design.**

Brad names the distinction this forces, and it is worth quoting because it is *our* distinction:

*Phase 2 asks a **concept-level** question — given a node's description, which lexicon concept does it represent? The collision finding asks an **instance-level** question — given two nodes correctly classified as the same concept, can the text tell you which occurrence you're looking at? Text structurally cannot, not because it's badly written but because it's byte-for-byte identical by construction. That information lives in the path (or another structural signal: parent context, containing list key, **when** / **augment** target), not in the prose.*

His conclusion: **definitional anchoring needs both** — definitional text to resolve *what kind of thing* a node is, and structural/path context to resolve *which one*. Treating text-only anchoring as sufficient would silently conflate distinct instances that share a grouping. This is a direct match to the reference's two roles in our framing.

2.4 Cross-corpus separability — lexical methods can't bridge, which is the whole point

Under TF-IDF (pure lexical overlap), **none** of the headline trap nodes find their true cross-corpus counterpart above threshold — IETF and TAPI use **almost entirely disjoint vocabulary for the same concepts** (IETF "termination point... a port or an interface" vs TAPI "NEPs belonging to / owned by this Node"). This is a *negative* result for lexical methods specifically, and it is exactly the gap an LLM's world knowledge — rather than surface token overlap — has to close. It is empirical motivation for cognition over matching.

(Brad adds a careful scoping note: Phase 2's *aggregate* name-only score is not a clean test of this cross-corpus claim, because the lexicon's synonym lists were deliberately built with vocabulary from *both* corpora, so name-only often has the bridge pre-built. The one row that genuinely isolates the claim is **owned-node-edge-point**, whose lexicon entry has IETF-only synonyms — see §3.5.)

3. Phase 2 — name-only vs definition-based binding

3.1 The experiment

A **12-entry lexicon** of concepts (LEX-001 ... LEX-012). A **gold standard** mapping YANG nodes to the correct lexicon entry (or to **NONE** where no entry fits). An LLM "binder" is asked, per node, which lexicon entry it is — in two modes:

- **name-only:** given the node name + the lexicon entries' names/synonyms.
- **definition-based:** given the node's actual **description** text.

Scored with precision / recall / F1 in the NER convention (a system must both *notice* a concept exists and *get its label right*; a confidently-wrong non-**NONE** guess counts as both a false positive and a false negative). Three scope filters are reported; the one Brad's headline numbers cite is **equivalence_plus_subsumption**, which scores on lexicon id and already folds in "subsumed-by" relations.

3.2 The blindness guarantee (this matters)

An earlier version of the study had the *same* agent author the gold standard and produce the "blind" predictions in one conversation — a real contamination risk. **This version replaces that with isolated tool-call invocations:** each prediction is a fresh model instance with no filesystem access and no visibility into the gold standard or the conversation, given only the literal prompt text. 144 scored calls in the core dataset (72 candidates × 2 modes), plus extra diagnostic calls that are reported but not scored. **One call per decision, never re-rolled** — a rule that does real work later (see §3.4). This is a genuine blind evaluation, not a demo.

3.3 The number keeps moving — and why that's the honest part

The aggregate F1 comparison changed *every time Brad corrected his own instrument* (the gold standard and the lexicon), across six rounds:

Round	n	name-only F1	definition-based F1	What changed
Uncorrected	39	0.98	0.84	first blind run
Corrected	39	0.946	0.958	fixed 2 wrong gold rows + a lexicon boundary
Expanded	58	0.943	0.893	+ 19 rows; surfaced LEX-003 over-triggering
LEX-003 fixed	58	0.935	0.923	rewrote the over-triggering lexicon entry
Investigated	58	0.935	0.942	resolved a layer-protocol parent-context error
Final	72	0.923	0.945	+ 14 rows

The first-round headline (name-only *crushing* definition-based, 0.98 vs 0.84) was an **artefact of a flawed instrument**, not a real result: two gold rows were simply wrong, and the lexicon had a boundary problem that penalised the description reader. As the instrument was corrected, the ranking inverted; and — the key point — **as the sample grew (39 → 58 → 72) the definition-based lead widened rather than narrowed**. Brad's reading, which is correct: a result that gets *more* pronounced with more data is the signature of a real effect, whereas small-

n noise shrinks or flips. He is equally clear that six rounds of a moving number on a single-author gold standard means the *absolute* figures are still soft.

3.4 The finding that is actually robust: the *shape* of the errors, not the count

What is stable across every single round is **how each mode fails**, and the two modes fail in categorically different places.

Name-only always fails as a "false friend" — a literal name or synonym match that means something *different*, sometimes opposite, in context. The recurring cases, present in every round:

- `nsrlg` matches a "shared-risk-group" synonym but denotes the semantic **opposite** (link groups that must be kept mutually *disjoint*).
- `lifecycle-state` matches a synonym but names a **different** concept (provisioning lifecycle) than the intended administrative in/out-of-service state.
- `owned-node-edge-point` / `parent-node-edge-point` get pulled to the wrong entry by a surface cue with no real synonym overlap.
- The cleanest case of all: `/networks/network` **itself**, with no description available, name-only infers "a network contains nodes, therefore network is broader than node" from the bare path and name — a plausible, name-shaped inference that is simply wrong, and there is no description text in name-only mode to correct it.

Every name-only error, across every round, fits this one description: **a name, synonym, or name-shaped inference that turns out to be a false friend.**

Definition-based's remaining errors collapse to essentially one category: the source text is too terse to state the disambiguating fact. Its cleanest examples:

- `cep-list` — full description "The list of supported ConnectionEndPoint (CEP) instances." It **genuinely never says** whether the CEP faces the client or the network side of an adaptation, so no reasoning and no lexicon rewording can recover the binding.
- `client-svc`, `supporting-termination-point` — same shape: four-word or relationally-phrased descriptions that don't contain the needed fact.

This is a property of the source **text**, not a failure of method: when the description doesn't encode the fact, cognition can't manufacture it from the description alone, and would need something outside the text (live system state at lift time, or an authority decision). Definition-based fails *only* where there is nothing left in the description to reason from.

So: name-only fails where a name misleads; definition-based fails where the text runs out. The second is the respectable kind of error, and it marks the limit of what the published text alone can support. Whether each such case is a true frontier or a gap the lift would close is taken up in §5.

3.5 The three problems Brad fixed (and how he told fixable from irreducible)

Across the correction rounds he found and disclosed three distinct instrument flaws, and — this is the discipline worth noting — he kept them apart from each other and from noise:

1. **Two gold-standard rows were simply wrong** (`nsrlg`, `lifecycle-state`) — the human-authored labels contradicted the source text; the blind definition-based calls had actually answered correctly. Fixing gold converted two definition-based "errors" into correct answers with no re-test.

2. **Lexicon boundary / over-triggering problems** — LEX-002/003/007 had overlapping "boundary point" definitions; separately, LEX-003's definition text literally contained "TE tunnel" and "path computation", so it over-triggered on any node whose text mentioned those words even when it wasn't a tunnel/LSP endpoint. Rewriting the lexicon entries with explicit cross-references fixed these; re-tested via fresh isolated blind calls.
3. **A missing rule for attribute/template containers** — property-bag containers (e.g. `te-link-attributes`) were mis-scored; one explicit sentence added to the prompt resolved it.

And the honesty markers:

- **Selection-effect caveat, self-imposed.** The correction round only re-investigated rows where the binder *disagreed* with gold — a confirmation-bias risk (you find gold wrong only where it's convenient). He explicitly names this and runs a **symmetric re-audit of the 37 rows where definition-based already agreed with gold**, finding no lexicon id needed to change — which closes the selection gap.
- **Reported an unstable result rather than re-rolling it.** `client-svc` returned three *different* answers across three isolated calls. Per the one-call-per-decision rule, he kept the original call's result as recorded and *reported the instability itself* as further evidence that this description is too terse to bind reliably — instead of quietly re-running until a call landed on the "right" answer.
- **Distinguished single-call sampling noise from real defects.** `supporting-network` gave a hedged, internally-inconsistent answer once; a diagnostic re-check came back clean. He flags it as *likely noise on a genuinely borderline case*, does **not** substitute the nicer answer (again per the no-re-roll rule), and leaves it as a recorded error while saying so.
- **Revised three gold labels on convergent evidence, not his own say-so** — where both name-only and definition-based independently converged on the same alternative to his initial guess, he treated that convergence (not his authority) as reason to change gold, and documented the reasoning per row.

3.6 Where the final errors actually sit (72-row set)

- **name-only: 7 errors** (TP 60, FP 7, FN 3, TN 5). **All 7 are the false-friend pattern.**
- **definition-based: 5 errors** (TP 60, FP 4, FN 3, TN 7). 3 of the 5 are the terse-source limit (`supporting-termination-point`, `client-svc`, `cep-list`); the other two are a likely sampling-noise borderline (`supporting-network`) and one genuine out-of-scope edge-case judgement (`network-id` : does an identifier of a network inherit the network concept, or is it out of scope?).

Final F1: **name-only 0.923, definition-based 0.945**, gap wider than at 58 rows (0.935 vs 0.942) and much wider than the corrected 39-row near-tie (0.946 vs 0.958 had them close).

4. What is solid vs what is soft

Solid (carry these):

- The **error-shape asymmetry** — name-only = false friends, definition-based = source-text terseness — is stable across all six rounds and both expansions. This is the trustworthy finding, independent of the exact F1.
- **Phase 1's corpus-level quality numbers** (~44–45% intensional; IETF-tautology vs TAPI-boilerplate split) have *no blindness confound* — they don't depend on the gold standard — and Brad rightly calls this the more trustworthy half of the study on its own.

- The **instance-vs-concept** distinction and the **disjoint-cross-corpus-vocabulary** result are clean and well-argued.

Soft (flag if cited):

- Gold standard, lexicon, and Phase-1 rubric labels are all **single-agent** ($\kappa = 0.880$ and 0.852 are *self-consistency*, not human validation). Brad states this himself; it's the one remaining contamination risk after the binder was isolated.
- **n = 72**, and the "widening trend" rests on two expansion points ($39 \rightarrow 58 \rightarrow 72$) — a small base to extrapolate from.
- The lexicon's synonyms were drawn from **both** corpora, which **inflates name-only's aggregate score** (its bridges are pre-built). The true name-only-vs-definition gap in the wild is therefore probably **larger** than 0.923 -vs- 0.945 shows; the **owned-node-edge-point** row (IETF-only synonyms) is the clean case where name-only fails as predicted and definition-based succeeds.

5. Reconciling this earlier work and the continuing study

This section reads the earlier study against the framework the continuing study has established, and states how its results are best understood at this point.

The framework established by the ongoing study, in brief. The lift, the production of a self-standing semantic model from a system's own representation, is a cognitive act performed by an agent with access to more than the published text: the running system, its instances, its configuration and operational context. Portability is comprehension by a cognitive agent that holds only the lifted model. A thin shared reference carries concept-level anchors in two roles: it names what kind of thing a node is, and it supplies facts an inert or terse counterpart cannot. A residue of authority and underdetermination remains, to be referred to a person.

The points that follow set the earlier study against that framework, one element at a time.

The binder is the portability test, run on un-lifted models. The definition-based binder is the portability test made concrete: a fresh, isolated cognition given only a node's published description and a thin reference, asked to comprehend it well enough to bind it. The models it reads were never lifted. A YANG description is authored by hand, frequently tautological or the literal **"none"**, and never produced by a cognition whose task was to make the model stand on its own. The study therefore measures the portability of the *un-lifted* model, and the wall it meets is the portability deficit the lift exists to close.

The wall is mostly a missing-lift gap. The terse-source failures are the shape of that wall: the CEP whose description ("the list of supported ConnectionEndPoint instances") never states whether it faces the client or the network side. Most such failures are not the irreducible frontier. Which side an endpoint faces is recorded in the running system, and a native cognition reads it at lift time and carries it into the lifted model; the fact is missing from the text, not from the system. The binder fails because it is handed an un-lifted artefact, and the reference correctly does not carry the fact, since "this endpoint faces the client" is instance-specific and belongs in that system's lifted model rather than in a shared reference. This is Phase 1's instance-against-concept distinction on the producing side: instance-level facts into the lift, concept-level facts into the reference. The genuinely irreducible residue is the smaller set of facts recorded *nowhere* in the system, true authority or underdetermination calls. The strength of this reading depends on how much of the terse-source set is latent in the system rather than truly absent; the flagship cases, the CEP side and the layer-protocol parent-context case, are latent, and each case is checkable directly.

Placement is a hand-off, and the lift's value is measurable. Cognition is placed within a system once, as the lift, so that a consumer above the system, holding only the artefact and a thin reference, can comprehend it. The lift carries the omitted facts from live state into the self-standing model. The study measures the state before that step, which makes it a baseline. Its modes fall on one value chain of increasing cognitive investment: names, authored description, and the lifted model. Name-only against definition-based is measured here, 0.945 against 0.923 in F1 at 72 cases, the margin widening with scale; the lifted model is the next rung, above the description the binder reads. The distance between the description-based ceiling and lifted-model portability is a measurable quantity: what the lift adds.

The floor, from the other side. The name-only errors mark the bottom of that chain. Every one of name-only's seven residual errors at 72 cases is a false friend, a name or synonym that matches while the meaning differs; the clearest is `/networks/network` itself, bound wrongly from the bare path on the inference that a network contains nodes and so is broader than one. Across the two corpora the vocabulary is near-disjoint: no trap node finds its counterpart above the lexical-similarity threshold. Surface matching cannot bridge that, which is the direct case for cognition over lexical or ontology matching.

What the raw material demands. The audit quantifies why the machinery is needed. Across the seven modules, only about 44 to 45 percent of description statements are genuine definitions; the remainder are tautology on the IETF side and, on the TAPI side, frequently the literal `"none"`. An agent reconciling real-world models works much of the time against text that carries little meaning, which is where the lift, the reference, and pragmatic judgement earn their place.

How the results read now. At this point in the current work, the study is best read as a measurement of the portability of the *un-lifted* model: a baseline for the lift, a map of the facts a lift must supply, and a localization of the small residue no lift reaches. The name-only, definition-based, and lifted-model levels are three points on one value chain; the study fixes the first two, and the terse-source wall it meets marks the work the lift is defined to do rather than a hard limit on comprehension.

6. Possible enhancements to the master report and a future NMRG I-D, from this study, for consideration

These are secondary to the reading in Section 5 and follow from it: candidate enhancements, drawn specifically from this work, offered rather than committed to.

- 1. Installed-base grounding for the motivation.** The audit's corpus-level numbers give the report, and an eventual NMRG Internet-Draft, a concrete answer to "why is ad hoc reconciliation needed" and "why not simply consume the published model": across the seven IETF and TAPI modules, only about 44–45% of description statements are genuine definitions, the IETF residue largely tautology and the TAPI residue frequently the literal string `"none"`. Candidate: a short empirical paragraph in the motivation.
- 2. Cognition placement as a hand-off, centred on the lift.** Candidate: state the placement of cognition as the hand-off set out in Section 5 (cognition within a system once, as the lift, so that a consumer above the system needs only the artefact and a thin reference), with the installed base's defects read as a specification of what the lift must supply.
- 3. The error taxonomy as evidence for the reference's two roles.** The false-friend (a name misleads) against terse-source (the description is exhausted) split, with the instance-against-concept distinction (text resolves what-kind, path resolves which-one), corroborates the reference's two roles from an independent direction. Candidate: cite as external corroboration where the reference is introduced.

4. **The un-lifted model as a measurable baseline for the lift.** Candidate: adopt the binder's published-model result as the un-lifted baseline against which the lift's value is measured (Section 5), as a tightened scenario in the study or a worked example in the I-D.
5. **Cross-corpus disjointness as motivation for cognition over matching.** The near-disjoint IETF/TAPI vocabulary, with no trap node finding its counterpart above the lexical-similarity threshold, is a crisp argument for world knowledge over lexical alignment. Candidate: a one-line empirical anchor.
6. **Citability.** Several of these become load-bearing only if the study exists in a citable form: a companion draft, a tech report, or a repository with a DOI. Worth deciding before an I-D leans on it.

A caution carried from §4: lean on the error-shape result and the corpus-level audit numbers, which are the sturdy findings; treat the absolute F1 values as indicative rather than settled, given the single-author gold standard and $n = 72$.